



NCI Protocol #: 10323

Patient ID:

Date:

1 of 8

Patient Name:

Date Collected:

Biopsy Site: **Liver**

Sample ID:

Telephone:

Tumor Content (%): **20\_49**

Patient DOB:

Referring Physician:

Primary Diagnosis: **Metastatic colorectal adenocarcinoma**

MoCha ID:

Fax:

Preanalytics to include tumor enrichment as required is performed by Van Andel Research Institute; Grand Rapids; MI 49503

## Table of Contents

## Page

Variant Details

2

Biomarker Descriptions

2

Relevant Therapy Summary

3

Clinical Trials Summary

4

## Report Highlights

2 Relevant Biomarkers

0 Therapies Available

13 Clinical Trials

## Relevant Colorectal Cancer Findings

| Gene  | Finding          | Gene  | Finding       |
|-------|------------------|-------|---------------|
| BRAF  | None detected    | NTRK1 | None detected |
| ERBB2 | None detected    | NTRK2 | None detected |
| KRAS  | <b>KRAS G12S</b> | NTRK3 | None detected |
| NRAS  | None detected    | RET   | None detected |

## Relevant Biomarkers

| Tier | Genomic Alteration | Relevant Therapies<br>(In this cancer type) | Relevant Therapies<br>(In other cancer type) | Clinical Trials |
|------|--------------------|---------------------------------------------|----------------------------------------------|-----------------|
| IA   | KRAS G12S          | None                                        | None                                         | 11              |
| III  | ROS1 R1942W        | None                                        | None                                         | 2               |

Public data sources included in relevant therapies: FDA<sup>1</sup>, NCCN

Tier Reference: Li et al. Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists. J Mol Diagn. 2017 Jan;19(1):4-23.

**Alerts informed by public data sources:** Contraindicated, Resistance

KRAS G12S cetuximab<sup>1</sup>, panitumumab<sup>1</sup>

Public data sources included in alerts: FDA<sup>1</sup>, NCCN

## Prevalent cancer biomarkers without relevant evidence based on included data sources

RB1 C283Y

Shahanawaz Jiwani, MD, PhD, Laboratory Director | CLIA ID 21D2097127

**Disclaimer:** The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2023.06(006). The content of this report has not been evaluated or approved by FDA, EMA or other regulatory agencies.

Variant Details

DNA Sequence Variants

| Gene | Amino Acid Change | Coding    | Variant ID | Locus          | Allele Frequency | Coverage | Transcript  | Variant Effect |
|------|-------------------|-----------|------------|----------------|------------------|----------|-------------|----------------|
| ROS1 | p.(R1942W)        | c.5824C>T | .          | chr6:117641147 | 44.41%           | 1995     | NM_002944.2 | missense       |
| KRAS | p.(G12S)          | c.34G>A   | COSM517    | chr12:25398285 | 48.70%           | 2000     | NM_033360.3 | missense       |
| RB1  | p.(C283Y)         | c.848G>A  | .          | chr13:48937080 | 37.99%           | 1977     | NM_000321.2 | missense       |

Comments:

N/A

Report Signed By

| Name | Role | Date |
|------|------|------|
|      |      |      |

Biomarker Descriptions

KRAS (KRAS proto-oncogene, GTPase)

Background: The KRAS proto-oncogene encodes a GTPase that functions in signal transduction and is a member of the RAS superfamily which also includes NRAS and HRAS. RAS proteins mediate the transmission of growth signals from the cell surface to the nucleus via the PI3K/AKT/MTOR and RAS/RAF/MEK/ERK pathways, which regulate cell division, differentiation, and survival<sup>1,2,3</sup>.

Alterations and prevalence: Recurrent mutations in RAS oncogenes cause constitutive activation and are found in 20-30% of cancers. KRAS mutations are observed in up to 10-20% of uterine cancer, 30-35% of lung adenocarcinoma and colorectal cancer, and about 60% of pancreatic cancer<sup>4</sup>. The majority of KRAS mutations consist of point mutations occurring at G12, G13, and Q61<sup>4,5,6</sup>. Mutations at A59, K117, and A146 have also been observed but are less frequent<sup>7,8</sup>.

Potential relevance: The FDA has approved the small molecule inhibitors, sotorasib<sup>9</sup> (2021) and adagrasib<sup>10</sup> (2022), for the treatment of adult patients with KRAS G12C-mutated locally advanced or metastatic non-small cell lung cancer (NSCLC). The FDA has also granted breakthrough therapy designation (2022) to the KRAS G12C inhibitor, GDC-6036<sup>11</sup>, for KRAS G12C mutation in non-small cell lung cancer. The small molecular inhibitor, RO-5126766, was granted breakthrough designation (2021) alone for KRAS G12V mutant non-small cell lung cancer or in combination with defactinib, for KRAS mutant endometrial carcinoma and KRAS G12V mutant non-small cell lung cancer<sup>12</sup>. The PLK1 inhibitor, onvansertib<sup>13</sup>, was granted fast track designation (2020) in combination with bevacizumab and FOLFIRI for second-line treatment of patients with KRAS-mutated metastatic colorectal cancer (mCRC). Additionally, the SHP2 inhibitor, BBP-398<sup>14</sup> was granted fast track designation (2022) in combination with sotorasib for previously treated patients with KRAS G12C-mutated metastatic NSCLC. The EGFR antagonists, cetuximab<sup>15</sup> and panitumumab<sup>16</sup>, are contraindicated for treatment of colorectal cancer patients with KRAS mutations in exon 2 (codons 12 and 13), exon 3 (codons 59 and 61), and exon 4 (codons 117 and 146)<sup>8</sup>. Additionally, KRAS mutations are associated with poor prognosis in NSCLC<sup>17</sup>.

RB1 (RB transcriptional corepressor 1)

Background: The RB1 gene encodes the retinoblastoma protein (pRB), and is an early molecular hallmark of cancer. RB1 belongs to the family of pocket proteins that also includes p107 and p130, which play a crucial role in the cell proliferation, apoptosis, and differentiation<sup>18,19</sup>. RB1 is well characterized as a tumor suppressor gene that restrains cell cycle progression from G1 phase to S

Biomarker Descriptions (continued)

phase<sup>20</sup>. Specifically, RB1 binds and represses the E2F family of transcription factors that regulate the expression of genes involved in the G1/S cell cycle regulation<sup>18,19,21</sup>. Germline mutations in RB1 are associated with retinoblastoma (a rare childhood tumor) as well as other cancer types such as osteosarcoma, soft tissue sarcoma, and melanoma<sup>22</sup>.

Alterations and prevalence: Recurrent somatic alterations in RB1, including mutations and biallelic loss, lead to the inactivation of the RB1 protein. RB1 mutations are observed in urothelial carcinoma (approximately 16%), endometrial cancer (approximately 12%), and sarcomas (approximately 9%)<sup>7</sup>. Similarly, biallelic loss of RB1 is observed in sarcomas (approximately 13%), urothelial carcinoma (approximately 6%), and endometrial cancer (approximately 1%)<sup>7</sup>. Biallelic loss of the RB1 gene is also linked to the activation of chemotherapy-induced acute myeloid leukemia (AML) and acute lymphoblastic leukemia (ALL)<sup>23,24,25</sup>.

Potential relevance: Currently, there are no therapies approved for RB1 aberrations.

ROS1 (ROS proto-oncogene 1, receptor tyrosine kinase)

Background: The ROS1 gene encodes the ROS proto-oncogene receptor tyrosine kinase 1 which exhibits structural similarity to anaplastic lymphoma kinase (ALK)<sup>26,27</sup>. Like ALK, ROS1 is the target of recurrent chromosomal rearrangements that generate fusion proteins containing the intact ROS1 tyrosine kinase domain combined with numerous fusion partner genes<sup>28</sup>. ROS1 fusion kinases are constitutively activated and drive oncogenic transformation<sup>29</sup>.

Alterations and prevalence: ROS1 fusions occur in approximately 1-2% of patients with non-small cell lung cancer (NSCLC) and are also observed in cholangiocarcinoma, gastric cancer, ovarian cancer, and glioblastoma<sup>26,30,31,32,33,34</sup>.

Potential relevance: The tyrosine kinase inhibitor, entrectinib<sup>35</sup>, is approved (2019) for the treatment of ROS1 fusion positive metastatic NSCLC. Crizotinib<sup>36</sup>, originally approved for the treatment of ALK positive NSCLC (2011), is also approved (2016) for the treatment of ROS1 positive NSCLC<sup>37</sup>. Acquired resistance to crizotinib in ROS1 positive NSCLC is associated with kinase domain mutations S1986F/Y, G2032R, D2033N, and L2155S<sup>38,39,40</sup>. The ROS1 tyrosine kinase inhibitor, repotrectinib<sup>41</sup>, was granted fast track and breakthrough designations (2020) for ROS1 positive NSCLC. The ROS-1 inhibitor, taletrectinib<sup>42</sup>, was also granted breakthrough therapy designation (2022) for the treatment of adult patients with advanced or metastatic ROS1-positive non-small cell lung cancer (NSCLC) who are ROS1 tyrosine kinase inhibitor (TKI) treatment naïve or previously treated with crizotinib. Ceritinib is a second generation ALK inhibitor approved (2017) for ALK positive NSCLC that has also shown efficacy in ROS1 positive NSCLC. In a phase II study, ceritinib demonstrated systemic and intra-cranial activity with an objective response rate (ORR) of 62% in patients with advanced ROS1 positive NSCLC<sup>43</sup>. In addition to crizotinib and entrectinib, ceritinib is recommended for first-line treatment of ROS1-positive NSCLC<sup>44</sup>. Lorlatinib is a CNS-penetrant third-generation ALK and ROS1 inhibitor with preclinical activity against almost all known ALK and ROS1 resistance mutations<sup>45,46</sup>. Lorlatinib is currently FDA approved (2018) for ALK positive metastatic NSCLC. In a phase I study testing lorlatinib in advanced ROS1-positive NSCLC, objective response was observed in 6/12 (50%) of patients<sup>47</sup>. Lorlatinib is recommended for subsequent therapy in ROS1 fusion-positive NSCLC in patients who have progressed after treatment with crizotinib, entrectinib, or ceritinib<sup>44</sup>.

Relevant Therapy Summary

In this cancer type

In other cancer type

In this cancer type and other cancer types

No evidence

| KRAS G12S                  |     |      |                  |
|----------------------------|-----|------|------------------|
| Relevant Therapy           | FDA | NCCN | Clinical Trials* |
| bevacizumab, chemotherapy  | ×   | ×    | ● (III)          |
| fruquintinib, chemotherapy | ×   | ×    | ● (II)           |

\* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

In this cancer type

In other cancer type

In this cancer type and other cancer types

No evidence

| KRAS G12S (continued)                  |     |      |                  |
|----------------------------------------|-----|------|------------------|
| Relevant Therapy                       | FDA | NCCN | Clinical Trials* |
| onvansertib, bevacizumab, chemotherapy | ×   | ×    | ● (II)           |
| panitumumab, trametinib                | ×   | ×    | ● (II)           |
| regorafenib, trametinib                | ×   | ×    | ● (II)           |
| afatinib, selumetinib                  | ×   | ×    | ● (I/II)         |
| OKI-179, binimetinib                   | ×   | ×    | ● (I/II)         |
| ET0038                                 | ×   | ×    | ● (I)            |
| JAB-3312                               | ×   | ×    | ● (I)            |
| RMC-6236                               | ×   | ×    | ● (I)            |
| ZEN-3694, binimetinib                  | ×   | ×    | ● (I)            |

| ROS1 R1942W      |     |      |                  |
|------------------|-----|------|------------------|
| Relevant Therapy | FDA | NCCN | Clinical Trials* |
| repotrectinib    | ×   | ×    | ● (I/II)         |
| LZ-001           | ×   | ×    | ● (I)            |

\* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Clinical Trials Summary

| KRAS G12S   |                                                                                                                                                                                                                                       |       |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| NCT ID      | Title                                                                                                                                                                                                                                 | Phase |
| NCT05379985 | Phase I/Ib Multicenter Open-Label Study of RMC-6236 in Subjects With Advanced Solid Tumors Harboring Specific Mutations in KRAS                                                                                                       | I     |
| NCT04045496 | A Phase I, Multi-Center, Open-Label Study to Evaluate the Safety, Tolerability, Pharmacokinetics, and Preliminary Evidence of Antitumor Activity of JAB-3312 in Adult Patients With Advanced Solid Tumors                             | I     |
| NCT02885753 | Systemic Oxaliplatin or Intra-arterial Chemotherapy Combined With LV5FU2 +/- Irinotecan and an Target Therapy in First Line Treatment of Metastatic Colorectal Cancer Restricted to the Liver                                         | III   |
| NCT05593328 | A Phase II, Randomized, Open-label Study of Onvansertib in Combination With FOLFIRI and Bevacizumab Versus FOLFIRI and Bevacizumab for Second Line Treatment of Metastatic Colorectal Cancer in Patients With a KRAS or NRAS Mutation | II    |

Shahanawaz Jiwani, MD, PhD, Laboratory Director | CLIA ID 21D2097127

## Clinical Trials Summary (continued)

### KRAS G12S (continued)

| NCT ID      | Title                                                                                                                                                                                                                  | Phase |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| NCT03087071 | A Phase II Enrichment Study of Panitumumab as a Single Agent or in Combination With Trametinib in Anti-EGFR-Refractory Stage IV Colorectal Cancer Patients                                                             | II    |
| NCT02450656 | Phase I/II Study With the Combination of Afatinib and Selumetinib in Advanced KRAS Mutant Positive and PIK3CA Wildtype Non-small Cell Lung Cancer                                                                      | I/II  |
| NCT05004441 | Fruquintinib Combined With mFOLFOX6/FOLFIRI in First-line Treatment for Metastatic Colorectal Cancer:HCCSC C02 Trial                                                                                                   | II    |
| NCT05354843 | A Phase I, Open-Label, Multi-Center Dose Finding Study to Investigate the Safety, Pharmacokinetics, and Preliminary Efficacy of SHP2 Inhibitor ET0038 Monotherapy in Patients With Advanced Solid Tumors               | I     |
| NCT04116541 | MegaMOST - A Multicenter, Open-label, Biology Driven, Phase II Study Evaluating the Activity of Anti-cancer Treatments Targeting Tumor Molecular Alterations /Characteristics in Advanced / Metastatic Tumors.         | II    |
| NCT05111561 | A Phase I Study of ZEN003694 in Combination With Binimetinib in Solid Tumors With RAS Pathway Alterations and Triple Negative Breast Cancer                                                                            | I     |
| NCT05340621 | NAUTILUS: A Phase I b/II Study of OKI-179 Plus Binimetinib in Patients With Advanced Solid Tumors and Activating Mutations in the RAS Pathway (Phase 1b) and in Patients With Advanced NRAS-Mutated Melanoma (Phase 2) | I/II  |

### ROS1 R1942W

| NCT ID      | Title                                                                                                                                                                                                                                    | Phase |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| NCT04094610 | A Phase I/II, Open-Label, Safety, Tolerability, Pharmacokinetics, and Anti-Tumor Activity Study of Repotrectinib in Pediatric and Young Adult Subjects With Advanced or Metastatic Malignancies Harboring ALK, ROS1, NTRK1-3 Alterations | I/II  |
| NCT05769075 | A Phase I, Multicenter, Open-label Study of TY-2136b, Administered Orally in Patients With Advanced or Metastatic Solid Tumors Harboring ALK, ROS1 or NTRK1-3 Alterations                                                                | I     |

## Assay Information

**Methodology, Scope, and Application:** The OCAv3 next-generation sequencing (NGS) assay identifies more than 3000 annotated mutations of interest (MOIs) broadly categorized into 5 mutation types: single nucleotide variants (SNVs), small insertions/deletions (Indels), large (>3 bases) insertions/deletions (Large Indels), copy number variants (CNVs), and gene fusions. The assay utilizes the Thermo Fisher Scientific Oncomine® Comprehensive Assay v3.0 (OCAv3), a next-generation sequencing (NGS) assay that utilizes a multiplex polymerase chain reaction (PCR) with DNA and RNA extracted from formalin-fixed tissue for sequencing on the Ion S5 XL platform and analyzed by the current version of Torrent Suite Software and Ion Reporter. The OCAv3 NGS Assay currently can reliably identify the presence or absence of >3000 known mutations and polymorphisms in the 161 unique genes compared to the Human Reference Genome hg19, including the genes listed below. The OCAv3 assay is a laboratory developed test designed to find gene mutations within tumors (somatic mutations).

**Analytical Sensitivity and Specificity:** The OCAv3 NGS Assay has been determined to be suitably analytically sensitive and specific for the various types of abnormalities within its reportable range, demonstrating 97.18% sensitivity compared with orthogonal assay results

Shahanawaz Jiwani, MD, PhD, Laboratory Director | CLIA ID 21D2097127

**Disclaimer:** The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2023.06(006).

and 100% specificity for the 5 mutation types reported. 99.99% or greater reproducibility has been demonstrated. All quality measures for this assay were within defined assay parameters.

**Hereditary and Germline Mutations:** This assay examines tumor tissue only and does not examine normal (non-tumor) tissue. Mutations detected by the assay may be present only in the tumor, or in every cell of the body (including non-tumor cells). This test also cannot tell whether a potential germline mutation causes or will cause a hereditary cancer syndrome. If the patient's history includes any one or combination of features suggestive of a possible hereditary cancer predisposition (such as, cancer arising at a young age, especially a cancer of a type unusual in younger patients; a personal history of multiple different types of cancers; or a significant cancer history among blood relatives, especially but not exclusively of the same type as the patient's) it is recommended that the physician arrange for the patient to meet with a genetic counselor and, if warranted, undergo the appropriate genetic test on normal (i.e. non-tumor) tissue (blood or cells brushed from the oral surface of the cheek) to check for a germline abnormality, regardless of the results of this research study.

**Report Content:** Thermo Fisher Scientific's Ion Torrent OncoPrint Reporter software was used in the generation of this report. Software was developed and designed internally by Thermo Fisher Scientific. The analysis was based on the current version of OncoPrint Reporter. The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. FDA information was sourced from [www.fda.gov](http://www.fda.gov) and is current as of this version. NCCN information was sourced from [www.nccn.org](http://www.nccn.org) and reflects its current version. Clinical Trials information reflects the current version. For the most up-to-date information regarding a particular trial, search [www.clinicaltrials.gov](http://www.clinicaltrials.gov) by NCT ID or search local clinical trials authority website by local identifier listed in 'Other Identifiers.' Variants are reported according to HGVS nomenclature and classified following AMP/ASCO/CAP guidelines (Li et al. 2017). Based on the data sources selected, variants, therapies, and trials identified in this report are listed in order of potential clinical significance but not for predicted efficacy of the therapies.

**Genes assayed for hotspot mutations:**

AKT1, AKT2, AKT3, ALK, AR, ARAF, AXL, BRAF, BTK, CBL, CCND1, CDK4, CDK6, CHEK2, CSF1R, CTNNB1, DDR2, EGFR, ERBB2, ERBB3, ERBB4, ERCC2, ESR1, EZH2, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, FOXL2, GATA2, GNA11, GNAQ, GNAS, H3F3A, HIST1H3B, HNF1A, HRAS, IDH1, IDH2, JAK1, JAK2, JAK3, KDR, KIT, KNSTRN, KRAS, MAGOH, MAP2K1, MAP2K2, MAP2K4, MAPK1, MAX, MDM4, MED12, MET, MTOR, MYC, MYCN, MYD88, NFE2L2, NRAS, NTRK1, NTRK2, NTRK3, PDGFRA, PDGFRB, PIK3CA, PIK3CB, PPP2R1A, PTPN11, RAC1, RAF1, RET, RHEB, RHOA, ROS1, SF3B1, SMAD4, SMO, SPOP, SRC, STAT3, TERT, TOP1, U2AF1, XPO1

**Genes assayed with full exon coverage:**

ARID1A, ATM, ATR, ATRX, BAP1, BRCA1, BRCA2, CDK12, CDKN1B, CDKN2A, CDKN2B, CHEK1, CREBBP, FANCA, FANCD2, FANCI, FBXW7, MLH1, MRE11, MSH2, MSH6, NBN, NF1, NF2, NOTCH1, NOTCH2, NOTCH3, PALB2, PIK3R1, PMS2, POLE, PTCH1, PTEN, RAD50, RAD51, RAD51B, RAD51C, RAD51D, RB1, RNF43, SETD2, SLX4, SMARCA4, SMARCB1, STK11, TP53, TSC1, TSC2

**Genes assayed for copy number variations:**

AKT1, AKT2, AKT3, ALK, AR, AXL, BRAF, CCND1, CCND2, CCND3, CCNE1, CDK2, CDK4, CDK6, EGFR, ERBB2, ESR1, FGF19, FGF3, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, IGF1R, KIT, KRAS, MDM2, MDM4, MET, MYC, MYCL, MYCN, NTRK1, NTRK2, NTRK3, PDGFRA, PDGFRB, PIK3CA, PIK3CB, PPARG, RICTOR, TERT

**Genes assayed for detection of fusions:**

AKT2, ALK, AR, AXL, BRAF, BRCA1, BRCA2, CDKN2A, EGFR, ERBB2, ERBB4, ERG, ESR1, ETV1, ETV4, ETV5, FGFR1, FGFR2, FGFR3, FGR, FLT3, JAK2, KRAS, MDM4, MET, MYB, MYBL1, NF1, NOTCH1, NOTCH4, NRG1, NTRK1, NTRK2, NTRK3, NUTM1, PDGFRA, PDGFRB, PIK3CA, PPARG, PRKACA, PRKACB, PTEN, RAD51B, RAF1, RB1, RELA, RET, ROS1, RSPO2, RSPO3, TERT

## DISCLAIMER

This assay is considered a Laboratory Developed Test (LDT). Its performance characteristics have been determined through extensive testing although it has not been cleared by the US Food and Drug Administration and such approval is not required for clinical implementation. Furthermore, any comments included in the report are strictly interpretive and the opinion of the reviewer. This laboratory is certified under the Clinical Laboratory Improvements Amendments (CLIA) for the performance of high-complexity testing for clinical purposes. The correlative data presented is a result of the curation of published data sources, but may not be exhaustive. The content of this report has not been evaluated or approved by the FDA or other regulatory agencies.

## References

1. Pylayeva-Gupta et al. RAS oncogenes: weaving a tumorigenic web. *Nat. Rev. Cancer*. 2011 Oct 13;11(11):761-74. PMID: 21993244
2. Karnoub et al. Ras oncogenes: split personalities. *Nat. Rev. Mol. Cell Biol.* 2008 Jul;9(7):517-31. PMID: 18568040
3. Scott et al. Therapeutic Approaches to RAS Mutation. *Cancer J.* 2016 May-Jun;22(3):165-74. doi: 10.1097/PPO.000000000000187. PMID: 27341593
4. Weinstein et al. The Cancer Genome Atlas Pan-Cancer analysis project. *Nat. Genet.* 2013 Oct;45(10):1113-20. PMID: 24071849
5. Román et al. KRAS oncogene in non-small cell lung cancer: clinical perspectives on the treatment of an old target. *Mol Cancer*. 2018 Feb 19;17(1):33. doi: 10.1186/s12943-018-0789-x. PMID: 29455666
6. Dinu et al. Prognostic significance of KRAS gene mutations in colorectal cancer--preliminary study. *J Med Life*. 2014 Oct-Dec;7(4):581-7. PMID: 25713627
7. Cerami et al. The cBio cancer genomics portal: an open platform for exploring multidimensional cancer genomics data. *Cancer Discov*. 2012 May;2(5):401-4. PMID: 22588877
8. Allegra et al. Extended RAS Gene Mutation Testing in Metastatic Colorectal Carcinoma to Predict Response to Anti-Epidermal Growth Factor Receptor Monoclonal Antibody Therapy: American Society of Clinical Oncology Provisional Clinical Opinion Update 2015. *J. Clin. Oncol.* 2016 Jan 10;34(2):179-85. PMID: 26438111
9. [https://www.accessdata.fda.gov/drugsatfda\\_docs/label/2023/214665s004lbl.pdf](https://www.accessdata.fda.gov/drugsatfda_docs/label/2023/214665s004lbl.pdf)
10. [https://www.accessdata.fda.gov/drugsatfda\\_docs/label/2022/216340Orig1s000Corrected\\_lbl.pdf](https://www.accessdata.fda.gov/drugsatfda_docs/label/2022/216340Orig1s000Corrected_lbl.pdf)
11. <https://assets.cwp.roche.com/f/126832/x/5738a7538b/irp230202.pdf>
12. <https://investor.verastem.com//news-releases/news-release-details/verastem-oncology-reports-third-quarter-2022-financial-results>
13. <https://cardiffoncology.investorroom.com/2020-05-28-Cardiff-Oncology-Announces-Fast-Track-Designation-Granted-by-the-FDA-to-Onvansertib-for-Second-Line-Treatment-of-KRAS-Mutated-Colorectal-Cancer>
14. <https://bridgebio.com/news/bridgebio-pharma-announces-first-lung-cancer-patient-dosed-in-phase-1-2-trial-and-us-fda-fast-track-designation-for-shp2-inhibitor-bbp-398-in-combination-with-amgens-lumakras-sotorasib/>
15. [https://www.accessdata.fda.gov/drugsatfda\\_docs/label/2021/125084s279lbl.pdf](https://www.accessdata.fda.gov/drugsatfda_docs/label/2021/125084s279lbl.pdf)
16. [https://www.accessdata.fda.gov/drugsatfda\\_docs/label/2021/125147s210lbl.pdf](https://www.accessdata.fda.gov/drugsatfda_docs/label/2021/125147s210lbl.pdf)
17. Slebos et al. K-ras oncogene activation as a prognostic marker in adenocarcinoma of the lung. *N. Engl. J. Med.* 1990 Aug 30;323(9):561-5. PMID: 2199829
18. Korenjak et al. E2F-Rb complexes regulating transcription of genes important for differentiation and development. *Curr Opin Genet Dev*. 2005 Oct;15(5):520-7. doi: 10.1016/j.gde.2005.07.001. PMID: 16081278
19. Sachdeva et al. Understanding pRb: toward the necessary development of targeted treatments for retinoblastoma. *J. Clin. Invest.* 2012 Feb;122(2):425-34. PMID: 22293180
20. Dyson. RB1: a prototype tumor suppressor and an enigma. *Genes Dev*. 2016 Jul 1;30(13):1492-502. PMID: 27401552
21. Cobrinik. Pocket proteins and cell cycle control. *Oncogene*. 2005 Apr 18;24(17):2796-809. PMID: 15838516
22. Dommering et al. RB1 mutations and second primary malignancies after hereditary retinoblastoma. *Fam. Cancer*. 2012 Jun;11(2):225-33. PMID: 22205104
23. Anasua et al. Acute lymphoblastic leukemia as second primary tumor in a patient with retinoblastoma. *Oman J Ophthalmol*. May-Aug 2016;9(2):116-8. PMID: 27433042
24. Tanaka et al. Frequent allelic loss of the RB, D13S319 and D13S25 locus in myeloid malignancies with deletion/translocation at 13q14 of chromosome 13, but not in lymphoid malignancies. *Leukemia*. 1999 Sep;13(9):1367-73. PMID: 10482987
25. Gombos et al. Secondary acute myelogenous leukemia in patients with retinoblastoma: is chemotherapy a factor?. *Ophthalmology*. 2007 Jul;114(7):1378-83. PMID: 17613328
26. Bergethon et al. ROS1 rearrangements define a unique molecular class of lung cancers. *J Clin Oncol*. 2012 Mar 10;30(8):863-70. doi: 10.1200/JCO.2011.35.6345. Epub 2012 Jan 3. PMID: 22215748



## References (continued)

27. Davare et al. Structural insight into selectivity and resistance profiles of ROS1 tyrosine kinase inhibitors. *Proc Natl Acad Sci U S A*. 2015 Sep 29;112(39):E5381-90. doi: 10.1073/pnas.1515281112. Epub 2015 Sep 8. PMID: 26372962
28. Kohno et al. Beyond ALK-RET, ROS1 and other oncogene fusions in lung cancer. *Transl Lung Cancer Res*. 2015 Apr;4(2):156-64. PMID: 25870798
29. Lin et al. Recent Advances in Targeting ROS1 in Lung Cancer. *J Thorac Oncol*. 2017 Nov;12(11):1611-1625. PMID: 28818606
30. Shaw et al. Crizotinib in ROS1-rearranged non-small-cell lung cancer. *N Engl J Med*. 2014 Nov 20;371(21):1963-71. doi: 10.1056/NEJMoa1406766. Epub 2014 Sep 27. PMID: 25264305
31. Gu et al. Survey of tyrosine kinase signaling reveals ROS kinase fusions in human cholangiocarcinoma. *PLoS ONE*. 2011 Jan 6;6(1):e15640. PMID: 21253578
32. Charest et al. Fusion of FIG to the receptor tyrosine kinase ROS in a glioblastoma with an interstitial del(6)(q21q21). *Genes Chromosomes Cancer*. 2003 May;37(1):58-71. PMID: 12661006
33. Birch et al. Chromosome 3 anomalies investigated by genome wide SNP analysis of benign, low malignant potential and low grade ovarian serous tumours. *PLoS ONE*. 2011;6(12):e28250. PMID: 22163003
34. Lee et al. Identification of ROS1 rearrangement in gastric adenocarcinoma. *Cancer*. 2013 May 1;119(9):1627-35. PMID: 23400546
35. [https://www.accessdata.fda.gov/drugsatfda\\_docs/label/2022/212725s006lbl.pdf](https://www.accessdata.fda.gov/drugsatfda_docs/label/2022/212725s006lbl.pdf)
36. [https://www.accessdata.fda.gov/drugsatfda\\_docs/label/2022/202570s033lbl.pdf](https://www.accessdata.fda.gov/drugsatfda_docs/label/2022/202570s033lbl.pdf)
37. Kazandjian et al. Benefit-Risk Summary of Crizotinib for the Treatment of Patients With ROS1 Alteration-Positive, Metastatic Non-Small Cell Lung Cancer. *Oncologist*. 2016 Aug;21(8):974-80. doi: 10.1634/theoncologist.2016-0101. Epub 2016 Jun 21. PMID: 27328934
38. Song et al. Molecular Changes Associated with Acquired Resistance to Crizotinib in ROS1-Rearranged Non-Small Cell Lung Cancer. *Clin Cancer Res*. 2015 May 15;21(10):2379-87. doi: 10.1158/1078-0432.CCR-14-1350. Epub 2015 Feb 16. PMID: 25688157
39. Drilon et al. A Novel Crizotinib-Resistant Solvent-Front Mutation Responsive to Cabozantinib Therapy in a Patient with ROS1-Rearranged Lung Cancer. *Clin Cancer Res*. 2016 May 15;22(10):2351-8. doi: 10.1158/1078-0432.CCR-15-2013. Epub 2015 Dec 16. PMID: 26673800
40. Facchinetti et al. Crizotinib-Resistant ROS1 Mutations Reveal a Predictive Kinase Inhibitor Sensitivity Model for ROS1- and ALK-Rearranged Lung Cancers. *Clin Cancer Res*. 2016 Dec 15;22(24):5983-5991. Epub 2016 Jul 11. PMID: 27401242
41. <https://ir.tptherapeutics.com/news-releases/news-release-details/turning-point-therapeutics-granted-fda-breakthrough-therapy>
42. <https://www.anhearttherapeutics.com/news/press-releases/080322/>
43. Lim et al. Open-Label, Multicenter, Phase II Study of Ceritinib in Patients With Non-Small-Cell Lung Cancer Harboring ROS1 Rearrangement. *J Clin Oncol*. 2017 Aug 10;35(23):2613-2618. doi: 10.1200/JCO.2016.71.3701. Epub 2017 May 18. PMID: 28520527
44. NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 3.2023]
45. Zou et al. PF-06463922, an ALK/ROS1 Inhibitor, Overcomes Resistance to First and Second Generation ALK Inhibitors in Preclinical Models. *Cancer Cell*. 2015 Jul 13;28(1):70-81. PMID: 26144315
46. Zou et al. PF-06463922 is a potent and selective next-generation ROS1/ALK inhibitor capable of blocking crizotinib-resistant ROS1 mutations. *Proc. Natl. Acad. Sci. U.S.A.* 2015 Mar 17;112(11):3493-8. PMID: 25733882
47. Shaw et al. Lorlatinib in non-small-cell lung cancer with ALK or ROS1 rearrangement: an international, multicentre, open-label, single-arm first-in-man phase 1 trial. *Lancet Oncol*. 2017 Dec;18(12):1590-1599. PMID: 29074098